

인공지능수학기초 (01)

Mid-term Exam

Oct. 29, 2024

1. (4+4 pts) 아래 연립 선형 방정식을 지정하는 방법으로 푸시오.

$$A = \begin{pmatrix} 1 & 2 & -3 \\ 2 & -1 & 1 \\ 4 & -1 & 1 \end{pmatrix} \& \underline{\mathbf{b}} = \begin{pmatrix} 9 \\ 0 \\ 4 \end{pmatrix}$$

(a) Gauss 소거법.

(b) Cramer's rule.

2. (4+4 pts) 다음 행렬을 보시오.

$$A = \begin{pmatrix} 2 & -4 & 0 & 2 & 1 \\ -4 & 2 & 1 & 2 & 3 \\ 1 & -2 & 1 & 4 & 4 \end{pmatrix}.$$

(a) A 의 rank를 구하시오.

(b) Null space의 basis를 구하시오.

3. (4+6 pts) 행렬 A 의 eigenvalue가 각각 3과 -2이고, 대응되는 eigenvector가 다음과 같다.

$$\underline{\mathbf{v}}_3 = \begin{pmatrix} 3 \\ 4 \end{pmatrix} \& \underline{\mathbf{v}}_{-2} = \begin{pmatrix} -4 \\ 3 \end{pmatrix}$$

(a) A 를 구하시오.

(b) $A^{10} \begin{pmatrix} 2 \\ 11 \end{pmatrix}$ 을 구하시오.

4. (4 pts) 다음 두 행렬이 서로 닮지 (similar) 않음을 보이시오.

$$A = \begin{pmatrix} 2 & 1 & 4 \\ 0 & 2 & 3 \\ 0 & 0 & 4 \end{pmatrix} \& B = \begin{pmatrix} 1 & 0 & 0 \\ -2 & 4 & 0 \\ 3 & 0 & 4 \end{pmatrix}$$

5. (3+3+4 pts) 다음 문장이 맞으면 증명하고, 틀리면 반례를 드시오.

(a) 두 벡터 $\underline{\mathbf{u}}$ 와 $\underline{\mathbf{v}}$ 가 선형 독립이면, $(\underline{\mathbf{u}} + \underline{\mathbf{v}})$ 와 $(\underline{\mathbf{u}} - \underline{\mathbf{v}})$ 도 서로 선형 독립이다.

(b) 3×3 크기의 행렬들 중에서 역행렬을 갖지 않는 행렬들의 집합은 벡터 공간을 이룬다.

(c) $(A_{adj})^{-1} = A_{adj}^{-1}$.

$$1. a) \underline{x} = \begin{pmatrix} 2 \\ 5 \\ 1 \end{pmatrix}$$

$$b) \det(A) = -2,$$

$$\det(A_1) = -4,$$

$$\det(A_2) = -10$$

$$\det(A_3) = -2$$

$$2. \begin{pmatrix} 2 & -4 & 0 & 2 & 1 \\ -4 & 2 & 1 & 2 & 3 \\ 1 & -2 & 1 & 4 & 4 \end{pmatrix}$$

$$\rightarrow \begin{pmatrix} 2 & -4 & 0 & 2 & 1 \\ 0 & -6 & 1 & 6 & 5 \\ 0 & 0 & 1 & 3 & 7/2 \end{pmatrix}$$

$$\rightarrow \begin{pmatrix} \triangle 1 & 0 & 0 & 0 & 0 \\ 0 & \triangle 1 & 0 & -0.5 & -0.25 \\ 0 & 0 & \triangle 1 & 3 & 3.5 \end{pmatrix}$$

$\lambda_1 \quad \lambda_2 \quad \lambda_3 \quad \lambda_4 \quad \lambda_5$

$$a) \text{rank}(A) = 3$$

$$b) x_4 = x, \quad x_5 = v$$

$$x_3 = -3x - 3.5v$$

$$x_2 = 0.5x + 0.25v$$

$$x_1 = 0$$

$$\begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{pmatrix} = \begin{pmatrix} 0 \\ 0.5x + 0.25v \\ -3x - 3.5v \\ x \\ v \end{pmatrix}$$

$$= x \begin{pmatrix} 0 \\ 0.5 \\ -3 \\ 1 \\ 0 \end{pmatrix} + v \begin{pmatrix} 0 \\ 0.25 \\ -3.5 \\ 0 \\ 1 \end{pmatrix}$$

$$\Rightarrow x \begin{pmatrix} 0 \\ 1 \\ -6 \\ 2 \\ 0 \end{pmatrix} + v \begin{pmatrix} 0 \\ 1 \\ -14 \\ 0 \\ 4 \end{pmatrix}$$

$\therefore$ basis of null space

$$\begin{pmatrix} 0 \\ 1 \\ -1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ -1 \\ 4 \end{pmatrix}$$

3. $A = S \Lambda S^{-1}$

$$a) = \begin{pmatrix} 3 & -4 \\ 4 & 3 \end{pmatrix} \begin{pmatrix} 3 & 0 \\ 0 & -2 \end{pmatrix} \begin{pmatrix} 3 & -4 \\ 4 & 3 \end{pmatrix}$$

$$= \frac{1}{5} \begin{pmatrix} -1 & 12 \\ 12 & 6 \end{pmatrix}$$

$$b) \begin{pmatrix} 2 \\ 11 \end{pmatrix} = \alpha \begin{pmatrix} 3 \\ 4 \end{pmatrix} + \beta \begin{pmatrix} -4 \\ 3 \end{pmatrix}$$

$$= 2 \begin{pmatrix} 3 \\ 4 \end{pmatrix} + \begin{pmatrix} -4 \\ 3 \end{pmatrix}$$

$$\therefore A^{10} \begin{pmatrix} 2 \\ 11 \end{pmatrix} = A^{10} \left(2 \begin{pmatrix} 3 \\ 4 \end{pmatrix} + \begin{pmatrix} -4 \\ 3 \end{pmatrix} \right)$$

$$= 3^{10} \times 2 \times \begin{pmatrix} 3 \\ 4 \end{pmatrix} + (-2)^{10} \begin{pmatrix} -4 \\ 3 \end{pmatrix}$$

$$= \begin{pmatrix} 6 \times 3^{10} - 4 \times 2^{10} \\ 2 \times 3^{10} + 3 \times 2^{10} \end{pmatrix}$$

4. A & B triangular matrix

$$\therefore \text{char}(A) = |A - \lambda I|$$

$$= (2-\lambda)(2-\lambda)(4-\lambda)$$

$$\text{char}(B) = |B - \lambda I|$$

$$= (1-\lambda)(4-\lambda)(4-\lambda)$$

$$\text{char}(A) \neq \text{char}(B)$$

$\therefore$ Not similar

5. a) ^{True} $\underline{u}$ & $\underline{v}$ linearly indep.

$$\rightarrow \text{i.e. } a_1 \underline{u} + a_2 \underline{v} = \underline{0}$$

only trivial soln. exists

$$b_1(\underline{u} + \underline{v}) + b_2(\underline{u} - \underline{v}) = \underline{0}$$

$$\rightarrow (b_1 + b_2)\underline{u} + (b_1 - b_2)\underline{v} = \underline{0}$$

since $\underline{u}$ & $\underline{v}$ are linearly indep.

$$b_1 + b_2 = 0 \quad \& \quad b_1 - b_2 = 0$$

$$\rightarrow b_1 = b_2 = 0$$

$\therefore (\underline{u} + \underline{v})$ & $(\underline{u} - \underline{v})$ linearly indep.

b) false

counter example,

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} + \begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

c) $AA^{\text{adj}} = \det(A) I.$

& $A^{-1} A^{-1 \text{adj}} = \det(A^{-1}) I$
 $= \frac{1}{\det(A)} I.$

$\rightarrow A = \det(A) (A^{\text{adj}})^{-1}.$

$\rightarrow A = \det(A) A^{-1 \text{adj}}$

$$\therefore (A_{adj})^{-1} = A^{-1}_{adj}$$